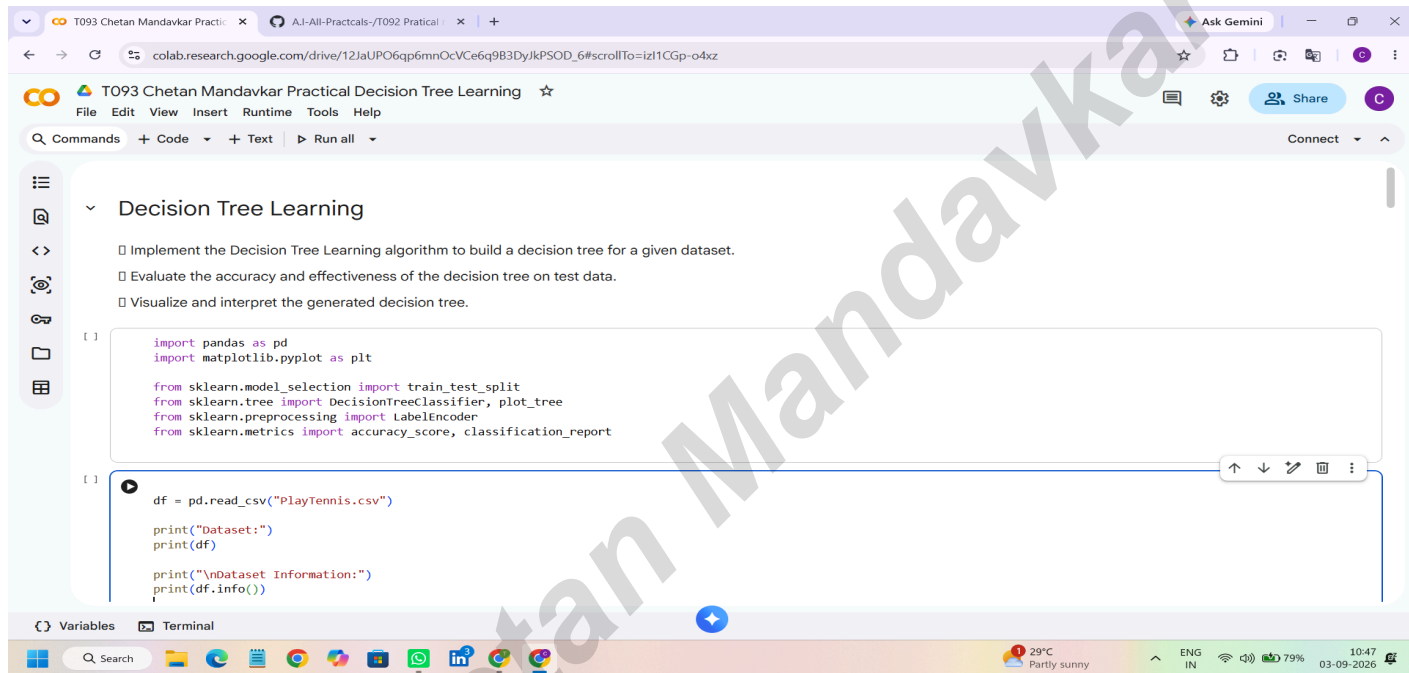


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Aim:- Decision Tree Learning

Implement the Decision Tree Learning algorithm to build a decision tree for a given dataset.

Evaluate the accuracy and effectiveness of the decision tree on test data. Visualize and interpret the generated decision tree.



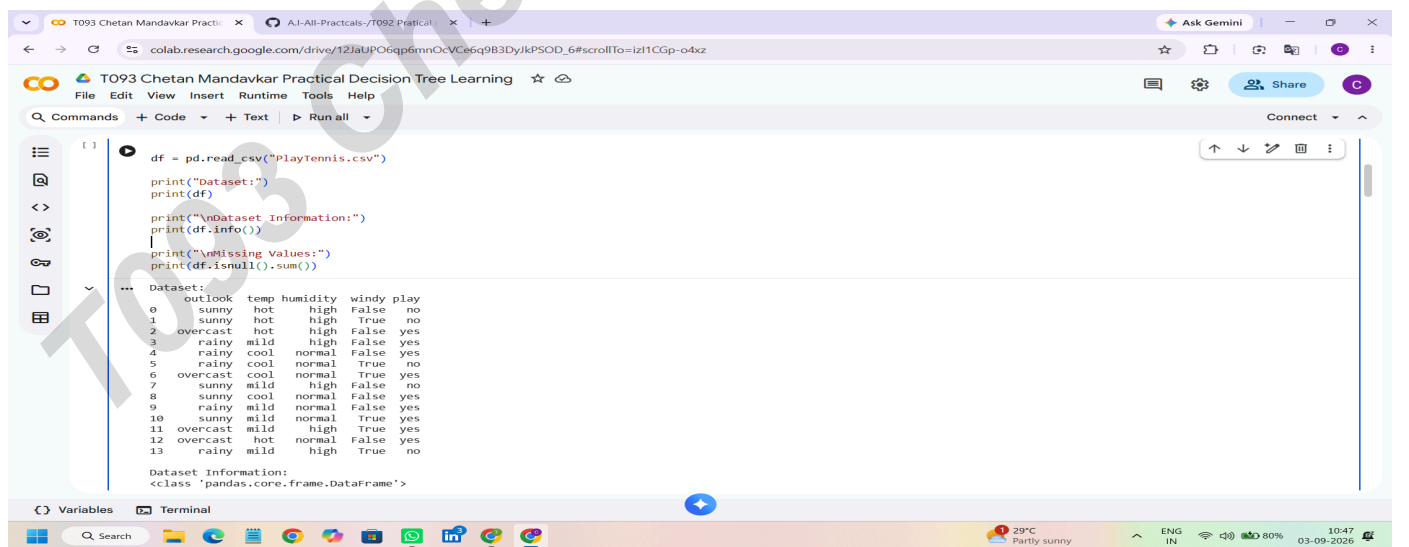
```
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import accuracy_score, classification_report

df = pd.read_csv("PlayTennis.csv")

print("Dataset:")
print(df)

print("\nDataset Information:")
print(df.info())
```



```
df = pd.read_csv("PlayTennis.csv")

print("Dataset:")
print(df)

print("\nDataset Information:")
print(df.info())

print("\nMissing Values:")
print(df.isnull().sum())
```

Dataset:

	outlook	temp	humidity	windy	play
0	sunny	hot	high	False	no
1	sunny	hot	high	True	no
2	overcast	hot	high	False	yes
3	rainy	mild	high	False	yes
4	rainy	cool	normal	False	yes
5	rainy	cool	normal	True	no
6	overcast	cool	normal	True	yes
7	sunny	mild	high	False	no
8	sunny	cool	normal	False	yes
9	rainy	mild	normal	False	yes
10	sunny	mild	normal	True	yes
11	overcast	mild	high	True	yes
12	overcast	hot	normal	False	yes
13	rainy	mild	high	True	no

Dataset Information:

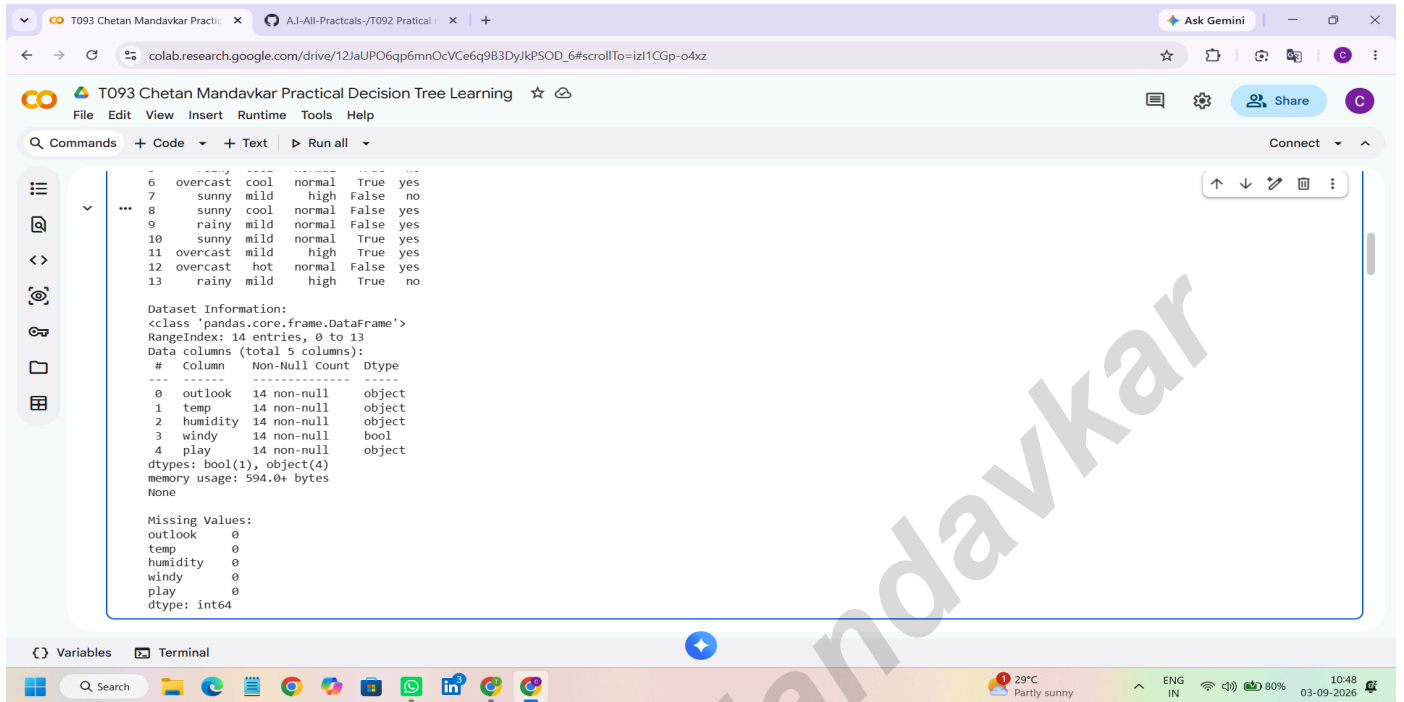
```
<class 'pandas.core.frame.DataFrame'>
```

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The screenshot shows a Google Colab environment with a Jupyter Notebook titled "T093 Chetan Mandavkar Practical Decision Tree Learning". The notebook contains a dataset of weather conditions and a decision tree model. The dataset is as follows:

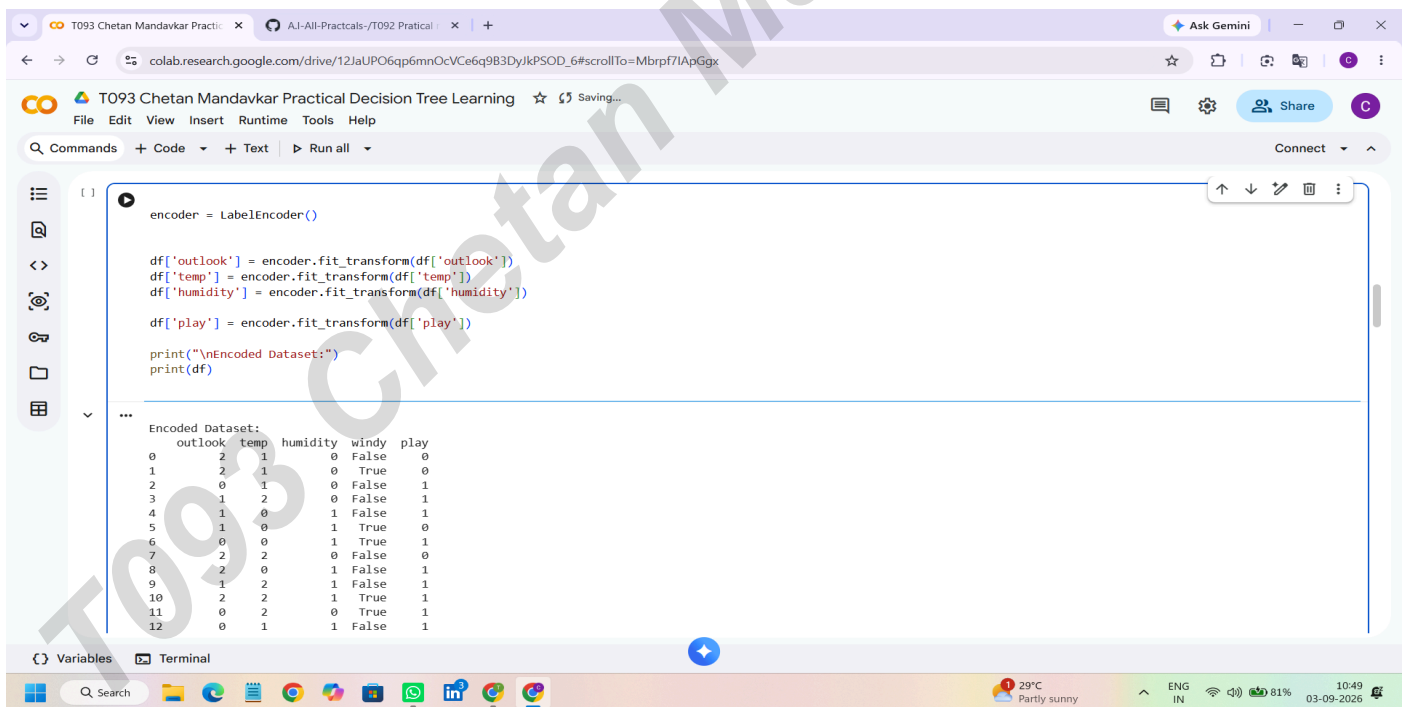
	outlook	temp	humidity	windy	play
6	overcast	cool	normal	True	yes
7	sunny	mild	high	False	no
8	sunny	cool	normal	False	yes
9	rainy	mild	normal	False	yes
10	sunny	mild	normal	True	yes
11	overcast	mild	high	True	yes
12	overcast	hot	normal	False	yes
13	rainy	mild	high	True	no

Dataset Information:

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 14 entries, 0 to 13  
Data columns (total 5 columns):  
#   column      Non-Null count  Dtype  
---  ---  
0    outlook    14 non-null         object  
1    temp       14 non-null         object  
2    humidity   14 non-null         object  
3    windy      14 non-null         bool  
4    play       14 non-null         object  
dtypes: bool(1), object(4)  
memory usage: 594.0+ bytes  
None
```

Missing Values:

```
outlook    0  
temp       0  
humidity    0  
windy       0  
play        0  
dtype: int64
```



The screenshot shows a Google Colab environment with a Jupyter Notebook titled "T093 Chetan Mandavkar Practical Decision Tree Learning". The notebook contains a decision tree model and its output. The code is as follows:

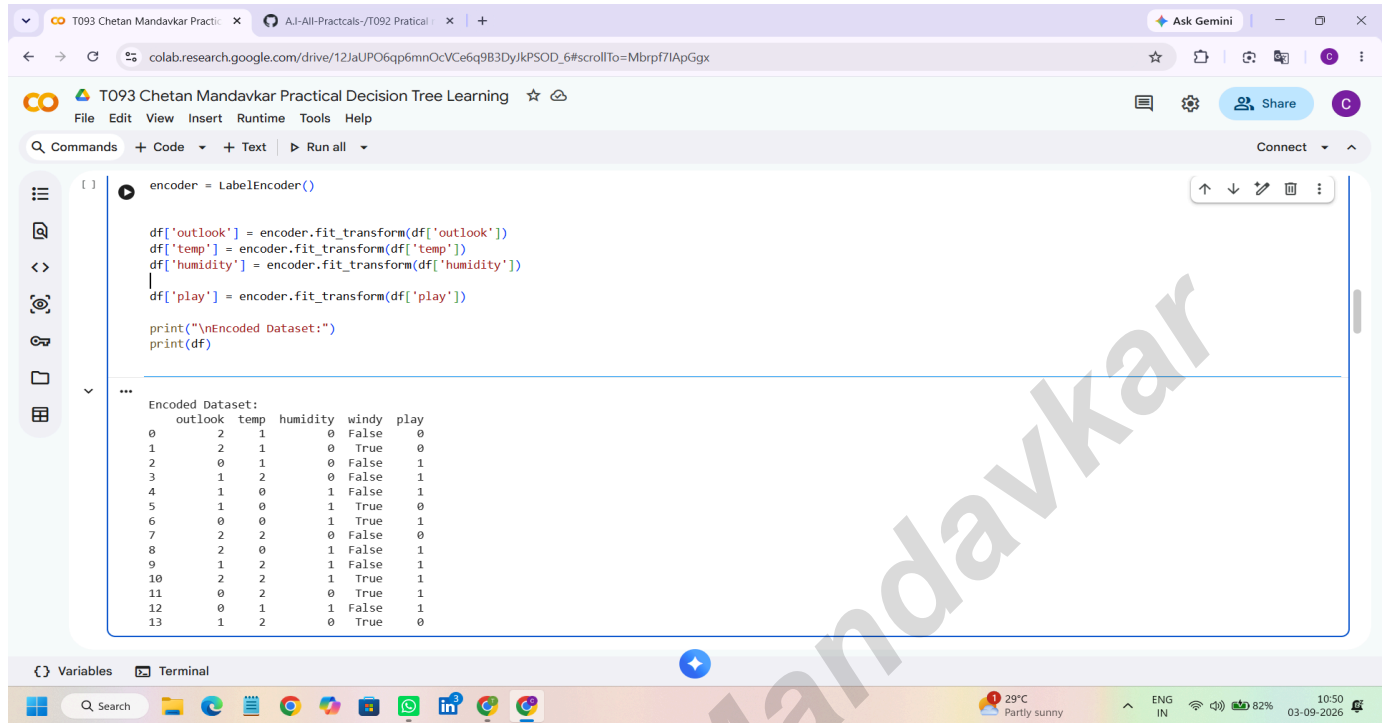
```
encoder = LabelEncoder()  
  
df['outlook'] = encoder.fit_transform(df['outlook'])  
df['temp'] = encoder.fit_transform(df['temp'])  
df['humidity'] = encoder.fit_transform(df['humidity'])  
  
df['play'] = encoder.fit_transform(df['play'])  
  
print("\nEncoded Dataset:")  
print(df)
```

Encoded Dataset:

	outlook	temp	humidity	windy	play
0	2	1	0	False	0
1	2	1	0	True	0
2	0	1	0	False	1
3	1	2	0	False	1
4	1	0	1	False	1
5	1	0	1	True	0
6	0	0	1	True	1
7	2	2	0	False	0
8	2	0	1	False	1
9	1	2	1	False	1
10	2	2	1	True	1
11	0	2	0	True	1
12	0	1	1	False	1

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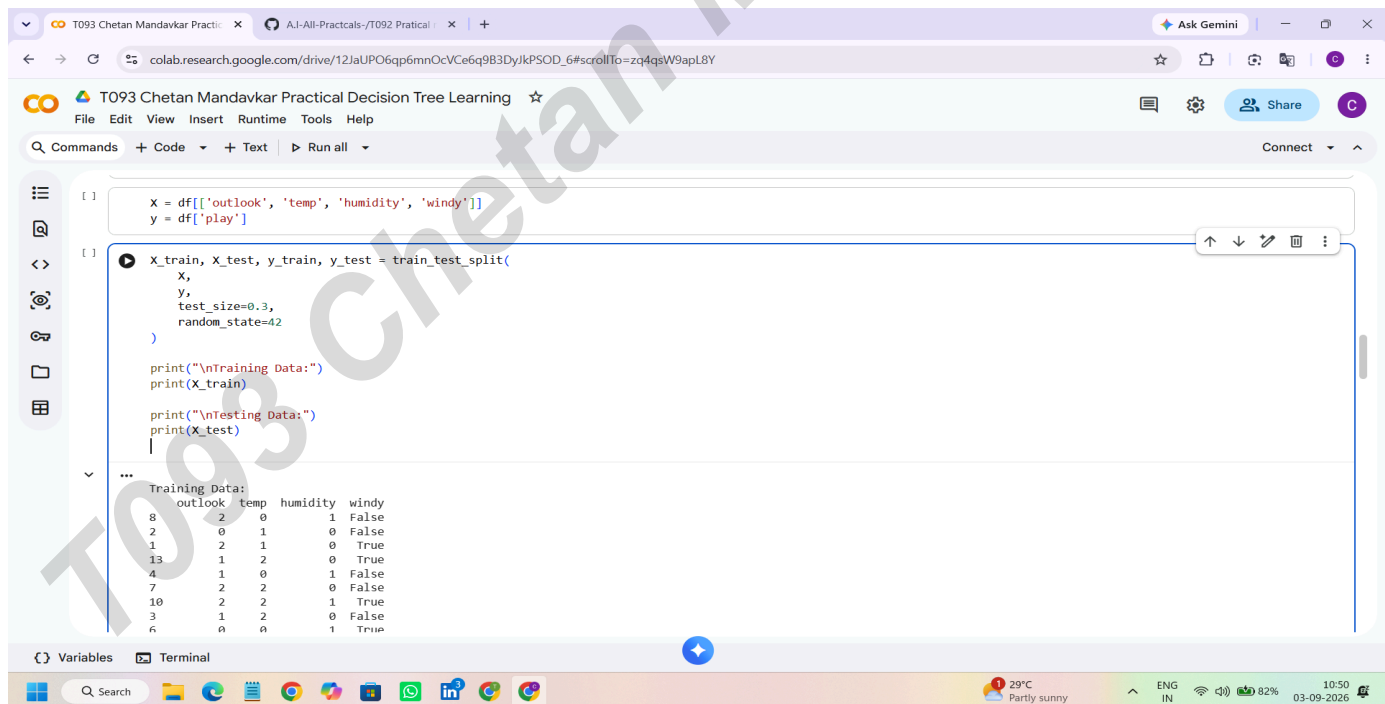
The screenshot shows a Google Colab notebook titled "T093 Chetan Mandavkar Practical Decision Tree Learning". The code in the first cell uses a `LabelEncoder` to transform categorical features: `outlook`, `temp`, `humidity`, and `play`. The output displays the "Encoded Dataset" as a DataFrame with 14 rows and 5 columns.

```
encoder = LabelEncoder()

df['outlook'] = encoder.fit_transform(df['outlook'])
df['temp'] = encoder.fit_transform(df['temp'])
df['humidity'] = encoder.fit_transform(df['humidity'])
df['play'] = encoder.fit_transform(df['play'])

print("\nEncoded Dataset:")
print(df)
```

	outlook	temp	humidity	windy	play
0	2	1	0	False	0
1	2	1	0	True	0
2	0	1	0	False	1
3	1	2	0	False	1
4	1	0	1	False	1
5	1	0	1	True	0
6	0	0	1	True	1
7	2	2	0	False	0
8	2	0	1	False	1
9	1	2	1	False	1
10	2	2	1	True	1
11	0	2	0	True	1
12	0	1	1	False	1
13	1	2	0	True	0



The screenshot shows the second cell of the Google Colab notebook. The code splits the dataset into training and testing sets using `train_test_split`. The output displays the "Training Data" as a DataFrame with 8 rows and 5 columns.

```
x = df[['outlook', 'temp', 'humidity', 'windy']]
y = df['play']

X_train, X_test, y_train, y_test = train_test_split(
    x,
    y,
    test_size=0.3,
    random_state=42
)

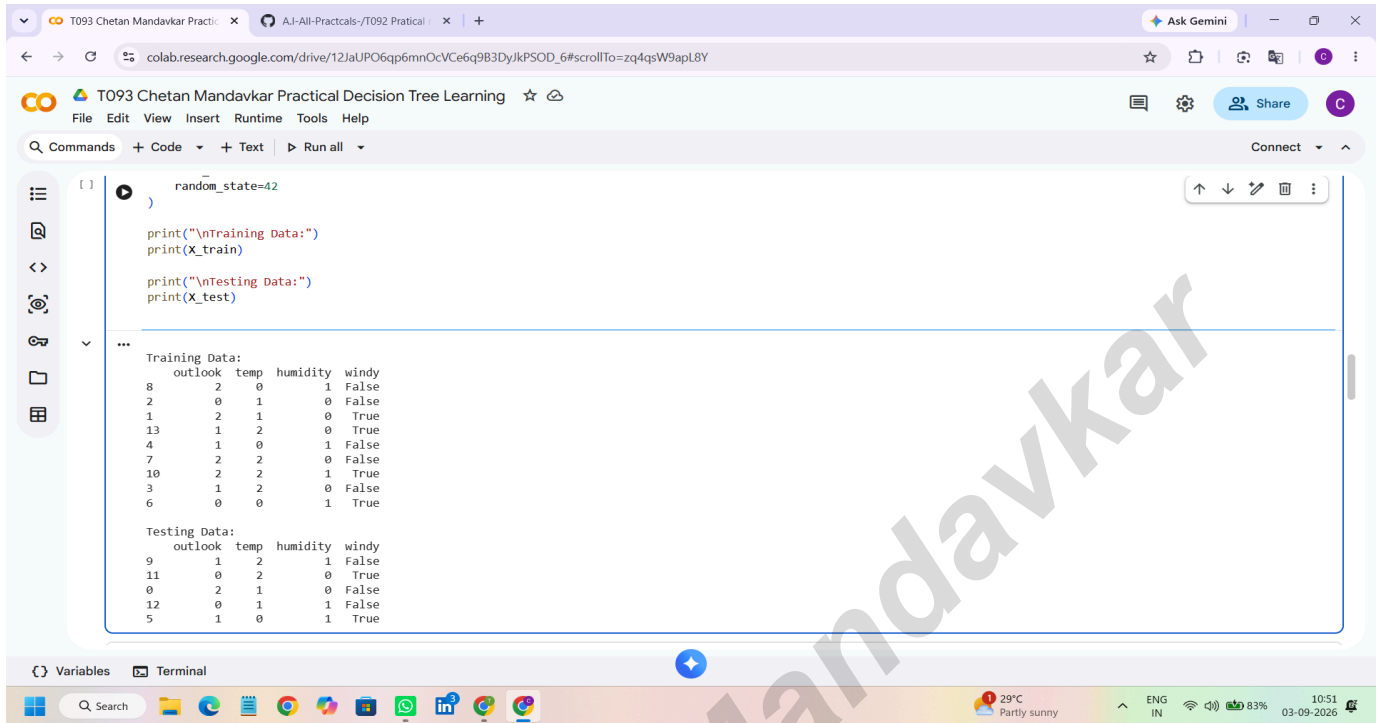
print("\nTraining Data:")
print(X_train)

print("\nTesting Data:")
print(X_test)
```

	outlook	temp	humidity	windy
8	2	0	1	False
2	0	1	0	False
1	2	1	0	True
13	1	2	0	True
4	1	0	1	False
7	2	2	0	False
10	2	2	1	True
3	1	2	0	False

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```
[ ] random_state=42
[ ] )

print("\nTraining Data:")
print(X_train)

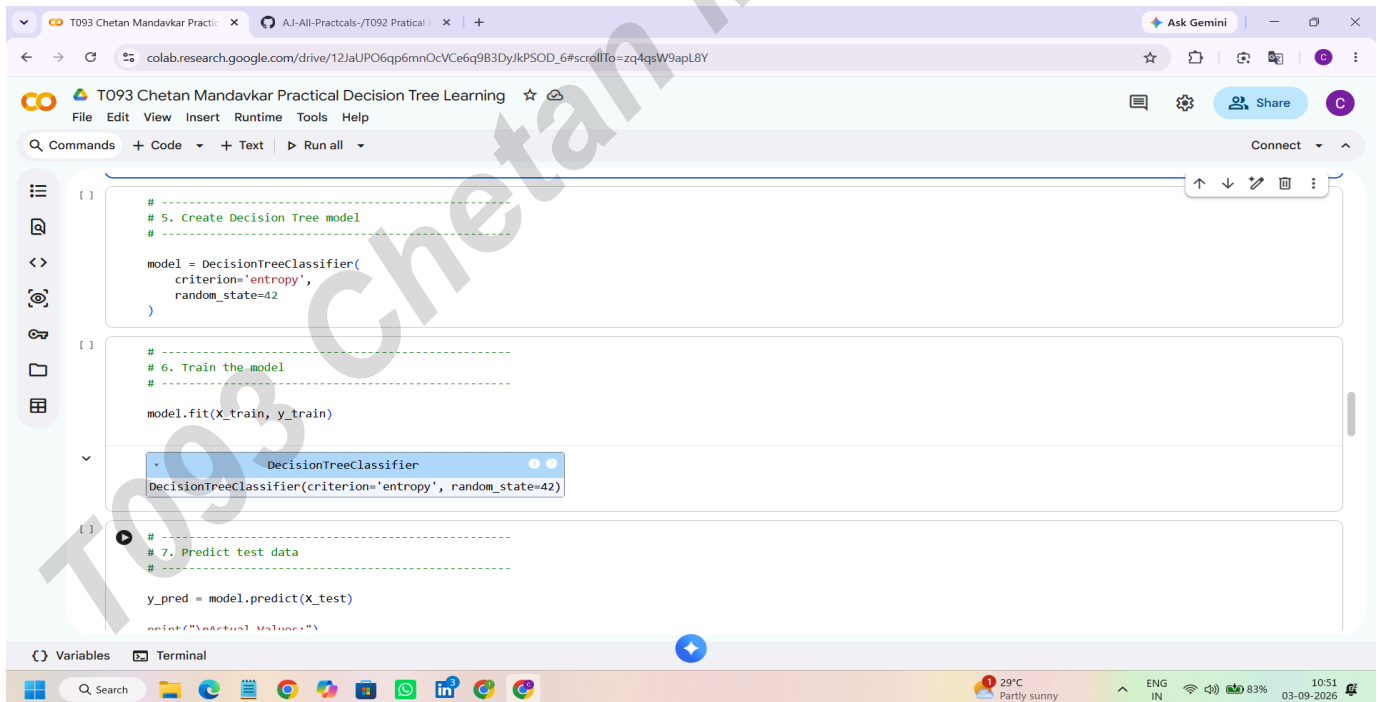
print("\nTesting Data:")
print(X_test)
```

Training Data:

	outlook	temp	humidity	windy	
8	2	0	1	False	
2	0	1	0	False	
1	2	1	0	True	
13	1	2	0	True	
4	1	0	1	False	
7	2	2	0	False	
10	2	2	1	True	
3	1	2	0	False	
6	0	0	1	True	

Testing Data:

	outlook	temp	humidity	windy	
9	1	2	1	False	
11	0	2	0	True	
0	2	1	0	False	
12	0	1	1	False	
5	1	0	1	True	



```
[ ] # -----
# 5. Create Decision Tree model
# -----

model = DecisionTreeClassifier(
    criterion='entropy',
    random_state=42
)

[ ] # -----
# 6. Train the model
# -----

model.fit(X_train, y_train)

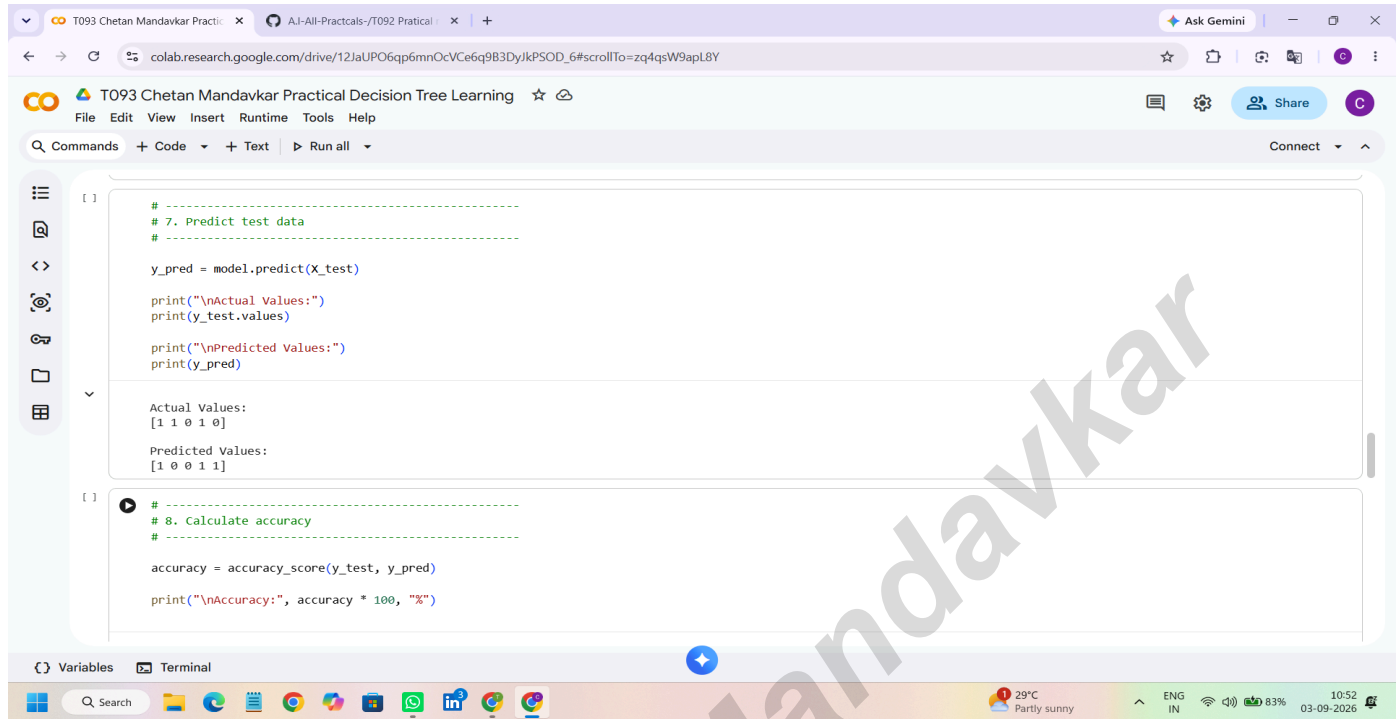
[ ] # -----
# 7. Predict test data
# -----

y_pred = model.predict(X_test)

print("\nActual Values:")
```

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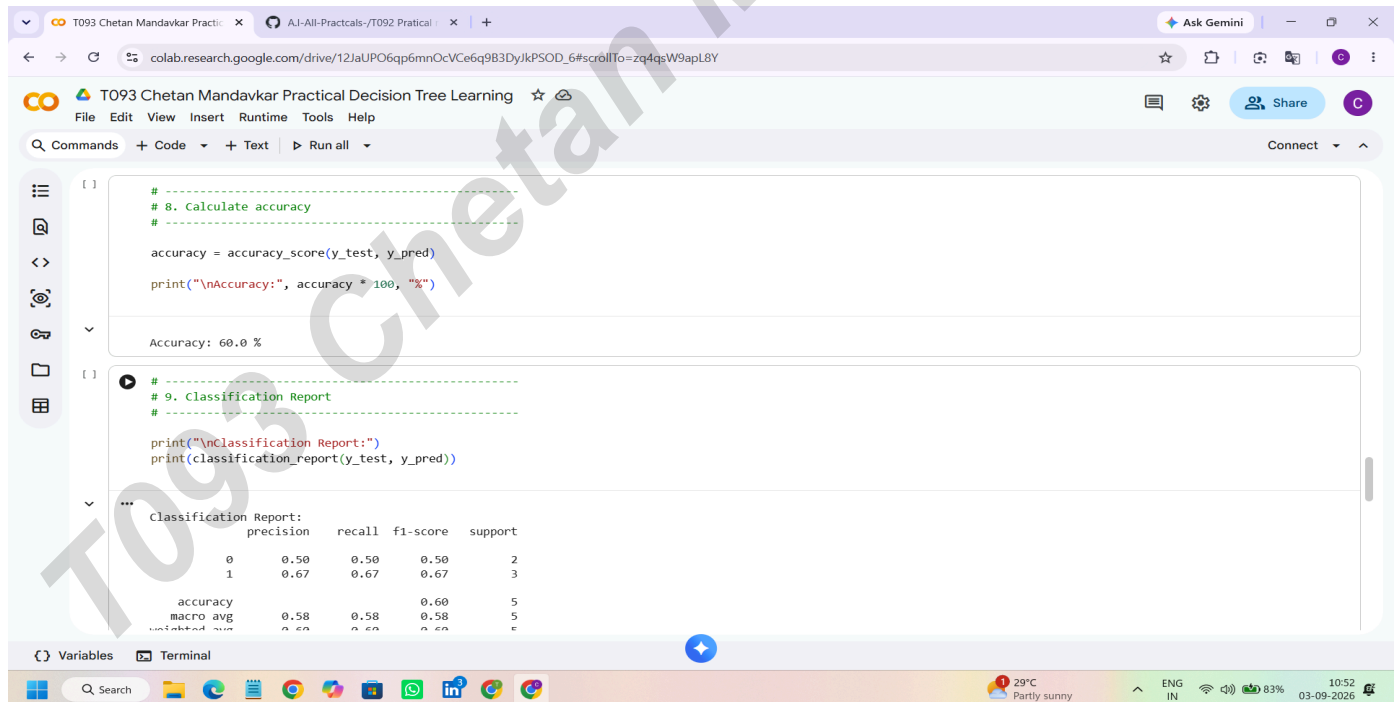


The screenshot shows a Google Colab environment with the following code and output:

```
# -----  
# 7. Predict test data  
# -----  
  
y_pred = model.predict(x_test)  
  
print("\nActual Values:")  
print(y_test.values)  
  
print("\nPredicted Values:")  
print(y_pred)  
  
Actual Values:  
[1 1 0 1 0]  
Predicted Values:  
[1 0 0 1 1]
```

```
# -----  
# 8. Calculate accuracy  
# -----  
  
accuracy = accuracy_score(y_test, y_pred)  
  
print("\nAccuracy:", accuracy * 100, "%")
```

The output shows the actual and predicted values for a test set, followed by the calculation of accuracy.



The screenshot shows the continuation of the Google Colab environment with the following code and output:

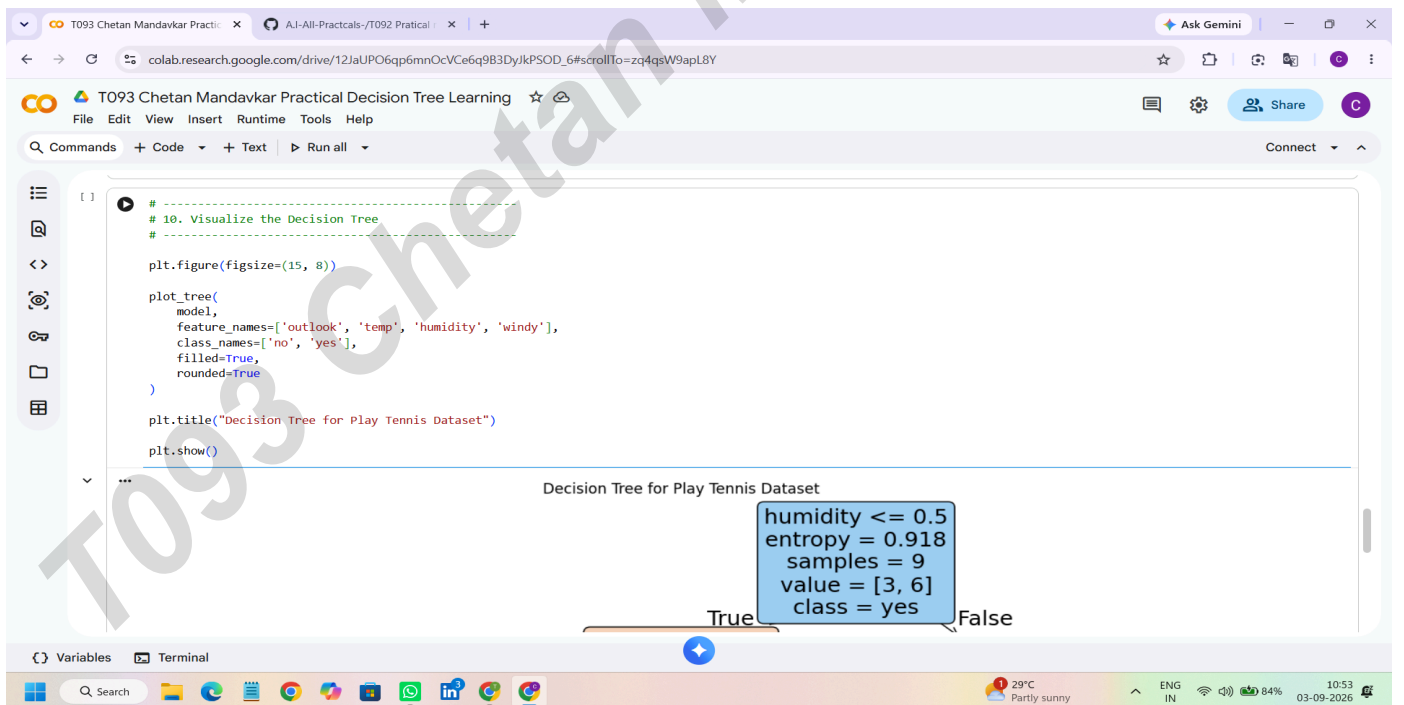
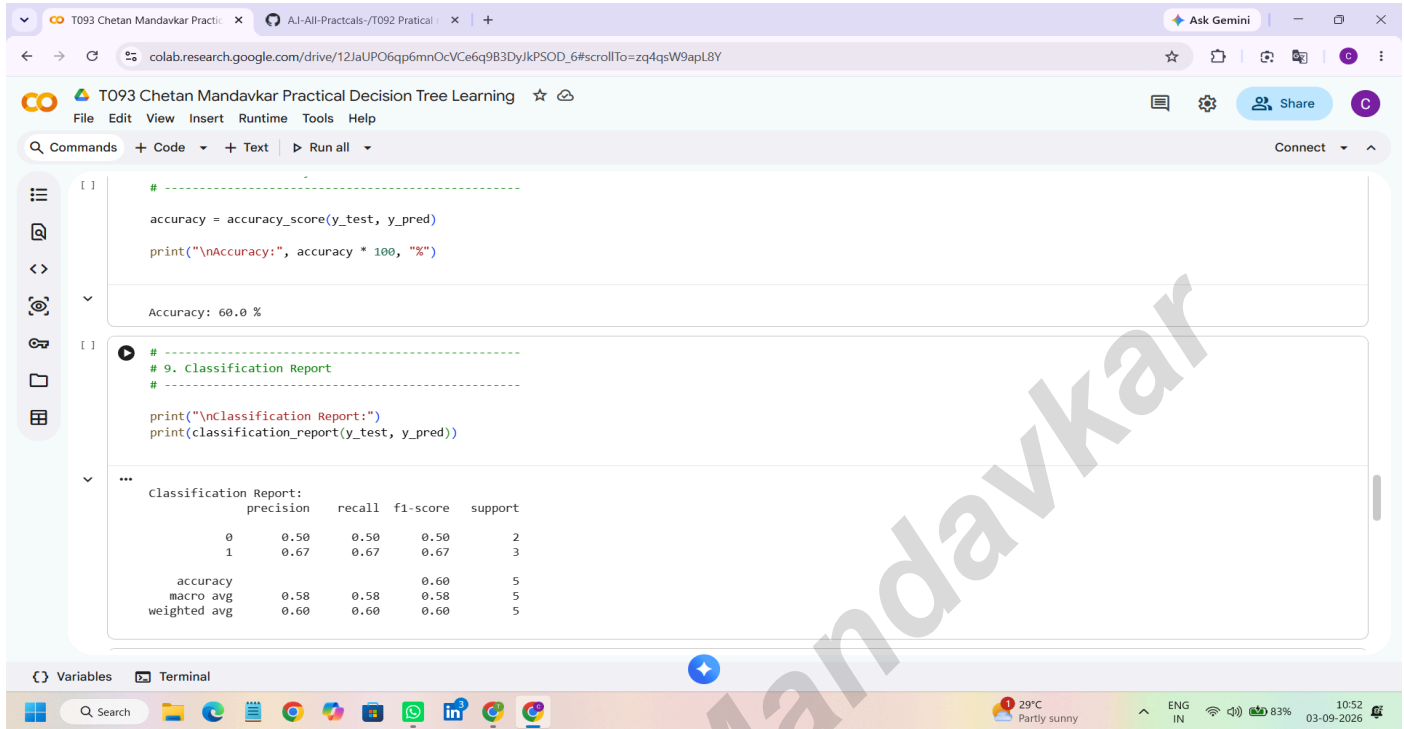
```
# -----  
# 8. Calculate accuracy  
# -----  
  
accuracy = accuracy_score(y_test, y_pred)  
  
print("\nAccuracy:", accuracy * 100, "%")  
  
Accuracy: 60.0 %
```

```
# -----  
# 9. Classification Report  
# -----  
  
print("\nClassification Report:")  
print(classification_report(y_test, y_pred))  
  
Classification Report:  
precision    recall  f1-score   support  
  
 0           0.50      0.50      0.50         2  
 1           0.67      0.67      0.67         3  
  
accuracy          0.58      0.58      0.60         5  
macro avg          0.58      0.58      0.58         5  
weighted avg          0.58      0.58      0.58         5
```

The output shows the accuracy of the model (60.0 %) and a detailed classification report.

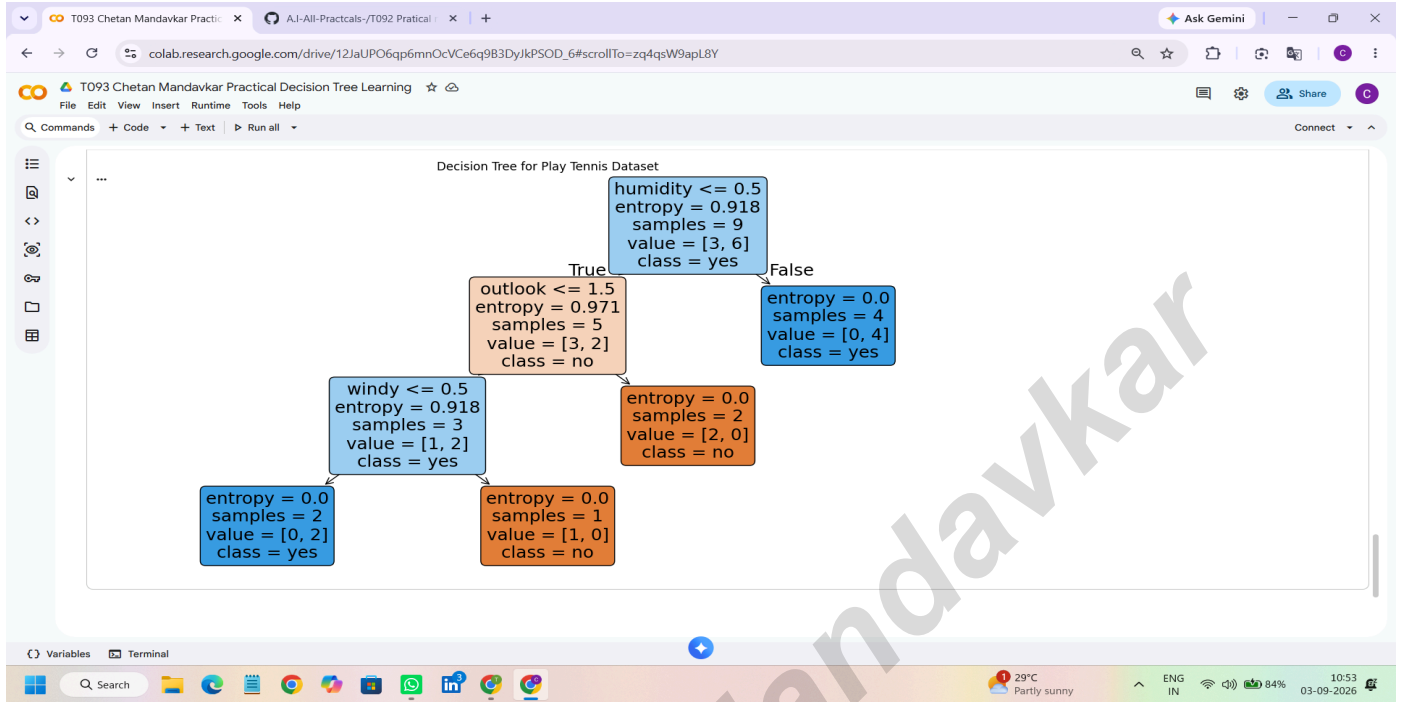
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